

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 6 Summer 2025 (Regular)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTCVOE605H - APPLICATIONS OF REMOTE SENSING AND GEOGRAPHIC INFORMATION SYSTEMS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 Which of the following best defines "remote sensing"?
 - a. Collecting data by physically touching the object
 - b. Collecting information about an object without being in direct contact
 - c. Only using satellites to collect data
 - d. Capturing images through a camera
- 2 Which part of the electromagnetic spectrum is commonly used in optical remote sensing?
 - a. X-rays
 - b. Ultraviolet
 - c. Visible and Infrared
 - d. Microwave
- 3 What does "spatial resolution" in remote sensing refer to?
 - a. The ability to generate 3D models
 - b. The distance a satellite can travel in one orbit
 - c. The smallest detail that can be detected by the sensor
 - d. The altitude of the satellite
- 4 Why might satellite remote sensing be preferred over ground surveying in highway planning?
 - a. It is less accurate
 - b. It covers smaller areas
 - c. It provides fast, large-area data collection
 - d. It is dependent on direct contact
- 5 What is the primary purpose of image enhancement in remote sensing?
 - a. To decrease spatial resolution
 - b. To increase cloud cover
 - c. To improve image interpretability
 - d. To compress file size

6 In the context of visual interpretation, “shadow” helps to identify which of the following features best?

- a. Soil texture
- b. Elevation of objects
- c. Water content
- d. Color variation

7 Which file format is commonly used for vector data?

- a. GeoTIFF
- b. Shapefile
- c. PNG
- d. CSV

8 What does GIS stand for?

- a. Global Imaging System
- b. Geographic Information System
- c. Geo-based Interaction Software
- d. General Image Storage

9 What is the main purpose of geo-referencing in GIS?

- a. To print maps on paper
- b. To increase map resolution
- c. To assign real-world coordinates to a digital image or map
- d. To compress spatial data files

10 What is the shape of the Earth typically approximated as in most geographic coordinate systems?

- a. Perfect sphere
- b. Flat plane
- c. Ellipsoid
- d. Cube

11 Which of the following is a datum used in GIS?

- a. Google Earth
- b. WGS 84
- c. ArcGIS
- d. Vector Map

12 Why are map projections necessary?

- a. To calculate internet speed
- b. To store raster files
- c. To represent the 3D Earth on 2D surfaces like maps
- d. To enhance satellite imagery

Q2. Solve the following.

- A) What are commonly used websites or platforms to download remote sensing data? 6
- B) Explain various important elements of visual interpretation of image Data with 6 neat labelled diagram.

Q3. Solve the following.

- A) Explain the ideal and real remote sensing systems in terms of all remote sensing concepts. 6
- B) Describe how LiDAR works and mention its key benefits and limitations. 6
- Q4. Solve Any Two of the following.
- A) Compare supervised and unsupervised classification techniques with advantages and limitations. 6
- B) Describe various thematic maps generated through remote sensing and civil engineering applications. 6
- C) Evaluate the applications of remote sensing and GIS in water resource and watershed management. 6
- Q5. Solve Any Two of the following.
- A) Compare raster and vector data formats with examples of their use in real world GIS applications. 6
- B) Describe the differences between georeferencing and geocoding in GIS applications. 6
- C) Write a detailed note on Geographic Information System with its advantages and limitations. 6
- Q6. Solve Any Two of the following.
- A) What is a datum? Explain different types of datum systems used in geodesy. 6
- B) Differentiate between DEM, DTM, and DSM with neat diagrams and applications. 6
- C) Explain the concept of geographic coordinate systems with the help of a diagram, including its strengths and weaknesses. 6

*** End ***